EFY TESTED
PREETI BHAKTA & SANI THEO

Getting Started With NodeMCU DEVELOPMENT BOARD For IoT

T.K. HAREENDRAN

Since Internet of Things (IoT) projects are now quite popular, I recently bought a NodeMCU board to try IoT application development. The NodeMCU is an open source development board based on ESP8266EX microcontroller with integrated Wi-Fi transceiver. Uploading programs to NodeMCU from any computer via microUSB port is very easy as it supports several programing languages. This makes NodeMCU a smart choice to play with the IoT. Here I explain how to get started with NodeMCU board using some common tools including Arduino, PuTTY, Lualoder and ESPlorer.

NodeMCU board comes in two popular versions: NodeMCU v0.9 with ESP12, and NodeMCU v1.0 with ESP12E (refer Fig. 1). NodeMCU v1.0 is the current version, which is not only breadboard-ready but also has some extra GPIOs (Fig. 2).

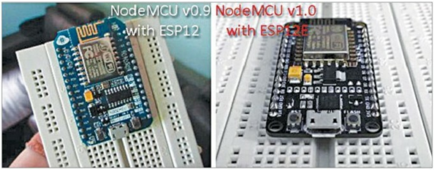

Fig. 1: NodeMCU versions

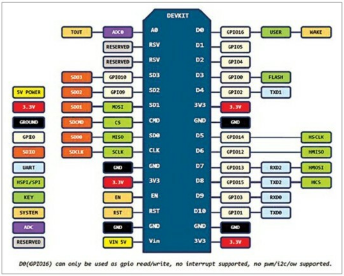

Fig. 2: NodeMCU v1.0 pinouts

First, run NodeMCU and Lua

Most NodeMCU boards come with updated Lua firmware for operating in Windows platform. If it's not included in your board, first refer to Firmware Flashing in the next section. Before using NodeMCU board, you need microUSB driver for Windows PC. Download driver from *http://www.silabs.com/products/mcu/Pages/USBtoUARTBridgeVCPDrivers.aspx*, and install it in your Windows PC. Next, connect NodeMCU board to the computer through USB interface. Fig. 3 shows 'Device Manager' window of Windows 7 (x64) showing connected port of NodeMCU as COM35 here.

Now, you can test the onboard LED in Lua shell. The on-board blue LED (GPIO16 –DO) signals execution of the procedure. Before this, also download and install the PuTTY terminal software in your computer (*http://www.chiark.greenend.org.uk/~sgtatham/putty/latest.html*)

With NodeMCU connected to your PC, run PuTTY to get into its Lua shell.

If NodeMCU is not detected by the software, press its reset (RST) button. Once NodeMCU is detected, enter the following code in the Lua shell window (refer Fig. 4) to test the onboard LED:

```
gpio.mode (0, gpio.OUTPUT)
gpio.write (0, gpio.HIGH)
print (gpio.read(0))
gpio.write (0, gpio.LOW)
print (gpio.read(0))
```

If everything is okay as per the hardware configuration, "gpio.write (0, gpio.HIGH)" turns off the LED, and "gpio.write (0, gpio.LOW)" turns on the LED. In the Lua shell window, the first line seems like some garbage data. This is usual, you just have to ignore that. Actually, these are random characters sent at a different baud rate but appear as garbage at normal rates.

NodeMCU with Arduino IDE

NodeMCU can be easily programmed through Arduino IDE